Ilya Tate

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Education

Arizona State University — Tempe, AZ
Bachelor of Science (B.S.) in Computer Science — *Expected May 2027*

- **Coursework:** Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management, Software Engineering

West-MEC Central Campus — Glendale, AZ
Full-Stack Development — *May 2022*

Experience

Project Lead — Sep 2026 – Present
Sun Devil Motorsports — *Tempe, AZ*

- Design the Vehicle Control Unit (VCU) and write its embedded Rust firmware on an STM32G4 microcontroller
- Implement 500 kbit/s CAN 2.0 messaging using async tasks and channels for frame input, processing, and output
- Improve reliability with a hardware watchdog and GitHub Actions CI that checks formatting and release builds
- Lead firmware work with an Agile Kanban board, pull requests, and an onboarding guide for new members
- Research and document PM100DX motor controller specs, wiring diagrams, and physical wiring
- Install and maintain mechanical and electrical components on the electric vehicle (EV)

Capstone Project Manager — Aug 2026 – Present
Arizona State University — *Tempe, AZ*

- Chosen by the sponsor and teammates to lead the 5-person team building Times Up, a mobile scheduling app
- Run Agile planning, review GitLab merge requests, and write most of the code and documentation
- Develop the Java, Spring Boot, and PostgreSQL REST API, with the database containerized in Docker Compose
- Modeled user accounts with Spring Data JPA, including age and email verification and BCrypt password hashing
- Set up the React Native and TypeScript frontend with Expo Router, including the app layout and calendar routes
- Built a CI/CD pipeline with EAS for preview and production builds, store submission, and over-the-air updates

Projects

Canplan | *JavaScript, Node.js, REST APIs, Chrome Extensions, Snowflake Cortex* — Oct 2026

- Co-built an AI Chrome extension at Hack Day x sunhacks 2026 that plans Canvas work using LLM estimates
- Wrote the Canvas REST API collector, which handles pagination and keeps only current, unsubmitted work
- Sped up syncs for long-history courses by fetching only open work in 3 parallel filtered requests
- Developed the planner's week view, date range filters, and live sync progress
- Wrote a mock Canvas server for end-to-end tests in a 54-test suite and added sync reports to find bottlenecks

Door Lock Monitor | *C++, Python, ESP32, IoT* — Jun 2024 – Aug 2024

- Built an IoT device that detects whether a deadbolt is locked using microphone audio and a magnetic sensor
- Programmed an ESP32 in C++ to read both inputs and report lock status over Wi-Fi
- Ran a local server in a home lab to receive and store lock status updates

Veterinary Adoption Site | *MongoDB, Express.js, React, Node.js, Next.js* — Feb 2022 – Apr 2022

- Co-built a MERN pet adoption site for West-MEC's veterinary program with animal listings and admin pages
- Added an admin role to the MongoDB user model for the site's admin pages

Pokemon Pokedex | *JavaScript, HTML, Bootstrap, Sass* — Dec 2021

- Wrote the JavaScript API integration for a static Pokedex web page built with a team of 3

Technical Skills

Languages: Java, Python, C/C++, Rust, JavaScript, TypeScript, HTML/CSS, SQL
Frameworks: Spring Boot, Spring Data JPA, React, React Native, Next.js, Expo, Node.js, Express.js, JUnit
Embedded: STM32, ESP32, Raspberry Pi, Embassy, CAN, SPI, UART
Practices: REST APIs, Agile, CI/CD, Automated Testing, Object-Oriented Programming (OOP)
Developer Tools: Git, GitHub, GitLab, GitHub Actions, Docker, PostgreSQL, MongoDB, Maven, Linux, Google Cloud